Intel Chiplet Market Signals

Proof-of-concept status report: Module 1 and Module 2

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1. Summary

- **Module 1 (Emerging Demand Radar)** - sample built and verified. 12 companies were screened; the deep dive covers **Cerebras, AWS and Tesla** (the brief's pilot set, with Cerebras as the named deep dive).
- **Module 2 (Competitive Movement Radar)** - foundation ready (competitor list, schema, event taxonomy, source checks), but its event pipelines have **not started** yet.
- **Scoring and shortlist (M1-7 / M2-7)** are **blocked** until the research team (Shu Qi) shares the scoring criteria, qualification rules and Intel mapping rulebook. Nothing is scored in the app until then.
- **Web app** - built and working locally: dashboard with charts, company views, evidence for every number, and the **Ask Intelligence** chatbot. Not yet deployed (waiting for GCP access or a Vercel login).
- **Data rule** - only source-verified records are shown: the exact sentence must be found in the source, and the source itself must state a date inside the last 12 months (from 1 Nov 2025).

Deep-dive companies - verified records in the app

197	132	65	271	51	379
Chip developments	Confirmed	Announced plans	Chip-engineering jobs	Partners & suppliers	Sources checked

Company	Developments	Confirmed	Jobs	Partners / suppliers	Main sources
Cerebras	92	58	54	17	IPO prospectus (424B4), two 10-Qs, cerebras.ai newsroom, trade press
AWS	79	62	138	22	Amazon 10-K / 10-Qs / earnings releases (AWS statements only), aboutamazon.com, amazon.jobs
Tesla	26	12	79	12	10-K and 10-Qs, ir.tesla.com, trade press, PredictLeads jobs

Full database (all 12 screened companies): 370 verified developments, 432 job postings, 103 relationships, 683 source documents.

2. How the work was done

Both modules share one backend, built first (tasks S1-S5), so Module 2 reuses the same companies, taxonomy, sources and extraction checks. Every pipeline follows the same five steps:

Step	What happens	Why it matters
1. Collect	Search official newsrooms, partner newsrooms, trusted trade press, SEC filings and job sites.	Official and primary sources come first; aggregators are excluded.
2. Store raw	Every provider response is saved untouched with its request and time (raw_provider_record).	Full audit trail; re-runs reuse saved responses, so credits are not spent twice.
3. Extract	Gemini (gemini-4-argon, temperature 0, fixed JSON schema) reads each document and returns structured records with an exact excerpt.	Same prompt and schema for every source; output is validated before use.
4. Check	Deterministic code checks each record: excerpt found word-for-word, company is in scope, date stated in the document and fully inside the window, evidence class by fixed rule.	The model is never the last word on a fact. Failures go to review sheets, not the app.
5. Publish	Accepted records are linked to their sources; duplicates from several outlets are merged into one record with extra sources kept as corroboration.	Every number in the app opens to its source sentence.

Evidence classes (fixed rule, not model opinion): *Confirmed* = something completed, stated by the company, a partner, a filing or press quoting the company. *Announced plan* = the same official sources, for something planned. Unattributed claims and analyst opinion are not published.

Module 1 - Emerging Demand Radar

- **Company master:** 22 companies (12 for Module 1, 11 for Module 2; AWS is in both), each field verified from Explorium, SEC filer records or official pages. Tricky mappings handled: AWS vs Amazon, Google vs Alphabet, Samsung Electronics vs Samsung Foundry.
- **Screening run:** 3 searches per company on official and trade-press domains for all 12 companies.
- **Hiring:** PredictLeads job history (Cerebras, Tesla, Tenstorrent) and amazon.jobs (AWS / Annapurna Labs); each posting classified into 8 chip-engineering role types from its own description. Vacancies are labelled as vacancies, not hires.
- **Deep dive (Cerebras, AWS, Tesla):** SEC filings read in full (424B4 prospectus, 10-K, 10-Q, earnings exhibits) plus 16 targeted searches per company (roadmap, foundry, packaging, suppliers, customers, funding, leadership, capacity, each chip programme). Amazon filings count for AWS only where the sentence names AWS or its chips.
- **Suppliers and partners:** taken only from named counterparties in verified records; no supplier is ever inferred.

Module 2 - Competitive Movement Radar

- **Ready:** competitor entities in the company master (Broadcom, Marvell, AMD, NVIDIA, Arm, TSMC, Samsung Foundry, plus hyperscaler programmes), 42 programmes, the six event types, the event table, and source checks for Broadcom, Marvell and TSMC (Exa, Google News, SEC filings for all competitor tickers).
- **Not started:** event discovery and extraction runs, pre-signal trail, the 5-10 representative events, and the Competitive Radar page content (the page currently shows an honest empty state). The Module 1 pipeline code is reusable for this.

3. Tools used

Area	Tool	Used for
Firmographics	Explorium API	Company match, domain, LinkedIn, HQ, business units (technographics dropped: generic IT stack)
Hiring	PredictLeads API, amazon.jobs	Job postings with posting and first-seen dates (Cerebras, Tesla, Tenstorrent; AWS via amazon.jobs)
News & web	Exa search API	Official newsrooms, partner newsrooms and an allow-list of trade press, with full text
Filings	SEC EDGAR (free)	Filing index and full documents: 10-K, 10-Q, 8-K exhibits, S-1 / 424B4
Fallback (probe only)	Google News RSS	Coverage check only - not used as evidence (redirect links, commentary-heavy)
AI extraction & chat	Google Gemini (gemini-4-argon)	Document extraction, job classification, Ask Intelligence answers (streamed, cited)
Database	SQLite / libSQL	One shared database: raw responses, sources, records, links; read-only snapshot ships with the app
Pipelines	Python 3.13: requests, BeautifulSoup, pydantic, google-genai, openpyxl, pytest	Collection, parsing, schema validation, review sheets, 32 automated tests
Web app	Next.js 16, React, Tailwind CSS, @libsql/client, @google/genai	Dashboard, charts, company and evidence pages, chatbot drawer, password gate
Engineering	Git / GitHub (BridgeAI-Tech/Intel-s-chiplet-business), Claude Code	Version control on branch feat/shared-foundation-s1-s3; development assistance

4. Task status

Shared foundation

Task	Status	Notes
S1 Database schema	Done	Shared IDs across both modules; event, publication and first-observed dates kept separate
S2 Taxonomy	Done	162 terms in 18 groups: role types, signal and event types, technologies, evidence classes, action tags
S3 Company master	Partly done	22 companies verified and exported; formal sign-off on the review sheet pending. Open: Cisco LinkedIn, Samsung Foundry HQ/domain
S4 Source validation	Done	All providers tested; findings in source_validation_report.md and source_probe_samples.xlsx
S5 Extraction layer	Done	Document and job-posting extraction, deterministic checks, dedup; run on real data
S6 App, RAG, validation	Partly done	App and Ask Intelligence done; deployment and full end-to-end validation pending

Module 1 - Emerging Demand Radar

Task	Status	Notes
M1-0 Scope	Done	12 screened; deep dive on Cerebras, AWS, Tesla
M1-1 Firmographics	Done	Verified domain, LinkedIn, HQ with sources
M1-2 Technographics	Done	Capability evidence from filings and news (Explorium tech stack not useful)
M1-3 Hiring	Done (3 companies)	AWS 138, Tesla 79, Cerebras 54 (+ Tenstorrent 161 in database). Other companies not collected
M1-4 News & filings	Done	197 developments for the 3 companies; 41 Cerebras records from the IPO prospectus and 10-Qs
M1-5 Suppliers	Done	51 named relationships for the 3 companies (TSMC, Arm, NVIDIA, Flex, Samsung...)
M1-6 Normalise & link	Done	Shared IDs; duplicate reports merged as corroborating sources
M1-7 Scoring & shortlist	Blocked	Needs Shu Qi's criteria, qualification rules and Intel rulebook
M1-8 UI	Done	Dashboard with 8 sections, charts, company pages, evidence, chatbot
M1-9 Validation	Partly done	Random samples checked by hand; strict publish rule. Brief section 12 questions not yet run as a formal test

Module 2 - Competitive Movement Radar

Task	Status	Notes
M2-0 Scope	Done	Competitor entities and value-chain roles in the company master (Arm kept as IP + Chiplet Alliance partner)
M2-1 Competitor profiles	Partly done	Profiles and programmes stored; Intel context partly filled
M2-2 Event discovery	Not started	Sources validated for Broadcom, Marvell, TSMC; pipeline to build (reuse Module 1 code)
M2-3 Extraction & classification	Not started	Six event types defined; extraction layer ready
M2-4 Pre-signal trail	Not started	Will link Module 1 hiring / supplier signals to events
M2-5 Representative events	Not started	5-10 events across 6+ entities, for Shu Qi review
M2-6 Event records	Not started	Table ready (m2_competitive_event)
M2-7 Intel implication mapping	Blocked	Needs Shu Qi's rulebook
M2-8 Competitive Radar page	Not started	Page shell exists; hidden from navigation for the client sample
M2-9 Validation	Not started	After M2-2 to M2-8

5. Review sheets shared

All sheets are in **data/review/** in the project repository. They hold what the reviewer needs to check, including everything the strict rules kept *out* of the app.

Sheet	Tabs (rows)	What it is / what to check
company_master_review.xlsx	Company master (22), Verification evidence (43), Domain checks (33), Programme master (42), Aliases (78)	The validated company list with the source behind each field. Reviewer marks sign-off; Programme master has reviewer_ok / comment columns.
source_probe_samples.xlsx	Explorium (3), PredictLeads silicon jobs (51), Exa (75), Google News (216), SEC filings (326)	Raw samples from each provider used to decide which sources to trust (task S4).
m1_news_review.xlsx	Not shown in app (756)	Screening-run extractions held back: subject not a tracked company (352), no in-window date in the document (111), unattributed claim (67), unparseable dates.
m1_deep_dive_review.xlsx	Not shown in app (906)	Deep-dive extractions held back: other company as subject (415), no in-window date (332), unattributed claim (25), unparseable dates.
m1_hiring_review.xlsx	Sheet (7)	Job postings that failed classification (6 errors) or whose excerpt was not found verbatim (1).

Run logs for each pipeline (m1_news_run.log, m1_hiring_run.log, m1_deep_dive_run.log) sit next to the sheets.

6. What is in the web app

A single dashboard for the three deep-dive companies. A company filter (All three / AWS / Cerebras / Tesla) applies to every section; every number opens to its source sentence.

Section	What the user sees
Overview	3-step how-to-read guide; 6 key numbers; company comparison bars; confirmed vs announced split; where the evidence comes from; activity month by month; activity types; partners named most; hiring skills; latest confirmed developments. With one company selected: a snapshot card (Intel relationship, latest development, top partner, top hiring area).
Developments	Cards for each development with the exact source sentence, date, source type and link; filter by type.
Hiring	Role mix per company, posting trend by month, recent openings with links to the job.
Partners & suppliers	Partners named most and every relationship with stage, evidence class, quote and source.
Intel relationship	Existing Intel ties per company with the quote that proves them.
SEC filings	Developments taken from 10-K, 10-Q, prospectus and earnings exhibits.
Scoring	States that scores appear only after the research team's criteria are agreed (no invented scores).
All sources	Every source document, filterable by type of source and website, with dates and links.
Where the data comes from	Providers used, stored responses and latest pipeline runs.
Ask Intelligence	Chat drawer on every page: answers only from verified records, with clickable citations and source list; focuses on the selected company; first words in ~3 s, full answer ~7 s.
Company profile pages	Verified profile fields (marked as source-verified), Intel relationship evidence, programmes, signals, hiring, suppliers.

Access and deployment: runs locally today; production build passes. Optional password gate (APP_PASSWORD). Recommended hosting: GCP Cloud Run (about $0-2 per month at POC traffic) once project access is granted; Vercel Pro is the quick alternative. Footer: Powered by BridgeAI.

7. Open items and next steps

#	Item	Owner
1	Share scoring criteria, qualification rules and Intel portfolio rulebook (unblocks M1-7 shortlist and M2-7)	Shu Qi
2	Sign off the company master review sheet	Research team
3	Grant GCP project access (Cloud Run, Artifact Registry, Cloud Build, Secret Manager, Logs Viewer) and deploy for the client	Manager / Satish
4	Build Module 2 pipelines, starting with Broadcom, Marvell, TSMC; select 5-10 representative events	Satish
5	Run the brief's section 12 example questions through Ask Intelligence as a formal test	Satish
6	Rotate the API keys used for the POC once the sample is signed off	Satish